Can This Experiment Detect Anticipatory Utility?

Identifiability and Power of a Delay-Manipulated Slot-Machine Task

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October 30, 2026

Abstract

Anticipatory utility (AU) holds that people value not only a reward but the experience of waiting for it. I designed a slot-machine task to test this behaviourally: participants choose between a sure amount (SAFE) and an equal-expected-value gamble (BET), and after a winning BET the payout arrives after a machine-specific delay of 8, 12 or 16 seconds. If savouring is real, longer delays should make BET more attractive even though the money is identical. Before collecting data, I ask whether this design can detect the effect at all.

Using simulated participants drawn from a reinforcement-learning model with and without an anticipatory term, I run parameter recovery, model recovery and a power analysis of the planned test. The original design is underpowered. With the planned pilot of 30 participants and a moderate savouring effect, the planned regression detects AU in 43% of simulated studies. Per-participant model comparison classifies a true savourer as AU only 42% of the time under AIC and 23% under BIC, and a quarter of fitted inverse temperatures hit their upper bound. A redesign that widens the delays to 2, 9 and 16 seconds and raises trials from 12 to 30 per machine reaches 95% power with the same 30 participants. No human data are used; the contribution is a design check that should precede data collection.

1 Introduction

Loewenstein [1987] proposed that the value of a delayed outcome includes the utility of anticipating it, which can make people prefer to delay pleasant outcomes. Iigaya et al. [2016] formalised savouring in a reinforcement-learning framework and showed that anticipation is boosted by the prediction error generated by an informative cue. Behaviourally isolating anticipatory utility is hard, because delay, probability and magnitude usually change together.

The task studied here holds expected value fixed and manipulates only the delay between a win cue and payment. I built it in Gorilla during a 2025 mentored MA research project at the Iigaya Lab; data collection did not take place before the project ended. This paper asks the question that should come first: *if anticipatory utility exists at a plausible strength, would the planned study detect it?* Following the recommendations of Wilson and Collins [2019], I answer it by simulating participants with known parameters and checking whether the planned analyses recover them.

2 The task

Each trial offers SAFE, a sure amount paid immediately, or BET, which pays twice that amount with probability 0.5 and nothing otherwise, so the two have equal expected value. A winning BET shows a win cue, then pays after the delay assigned to its machine: 8 s for machine A, 12 s for B and 16 s for C. A losing BET pays nothing immediately. The task has 36 main trials, 12 per machine, with stakes between \$4 and \$11. Two catch trials offer SAFE \$25 against a \$20 BET, where SAFE clearly dominates, to check attention. Trial order is randomised per participant. The full trial list is the Gorilla spreadsheet in the accompanying repository.

3 Models

Both models learn each machine's win probability from the outcomes of BET choices,

$$\hat{p}_m \leftarrow \hat{p}_m + \alpha\,(o_t - \hat{p}_m), \qquad \hat{p}_m^{(0)} = 0.5, \tag{1}$$

updated only after a BET on machine m, and choose with a logistic rule,

$$P(\mathrm{BET}_t) = \sigma(\beta\,[\,Q_{\mathrm{bet},t} - Q_{\mathrm{safe},t}\,])\,, \qquad Q_{\mathrm{safe},t} = S_t, \tag{2}$$

with amounts in units of \$10.

EV-RL. The value of betting is its learned expected value plus a general taste for gambling,

$$Q_{\mathrm{bet},t} = \hat{p}_m H_t + b, \tag{3}$$

where H_t is the winning amount. Parameters are (α, β, b).

AU-RL. A win is worth more when it can be savoured for longer,

$$Q_{\text{bet},t} = \hat{p}_m H_t \left(1 + \kappa \frac{D_m}{16\text{ s}}\right) + b, \tag{4}$$

where D_m is the machine's delay. The parameter κ is the extra value of a win, as a fraction of the prize, at the longest delay; $\kappa > 0$ is savouring and $\kappa < 0$ dread. The bias b shifts BET equally on every machine, so only κ can produce a preference that changes with delay. This is what makes the two models separable in principle.

4 Methods

Simulated participants. Each simulated participant draws $\alpha \sim U(0.05, 0.6)$, $\beta \sim U(1, 8)$ and $b \sim N(0, 0.3^2)$; AU participants also draw κ. These ranges are assumptions chosen to give choice behaviour that is neither deterministic nor random, not estimates from real data.

Fitting. Both models are fitted per participant by maximum likelihood with L-BFGS-B under box constraints, from five random starting points, keeping the best. Models are compared with AIC and BIC.

Analyses. (i) *Parameter recovery:* 200 AU participants with $\kappa \sim U(0, 1)$, fitted with AU-RL. (ii) *Model recovery:* 100 participants from each model, with $\kappa \sim U(0.3, 1)$ for AU, each fitted with both models. (iii) *Power:* for $N \in \{15, 30, 60, 120\}$ and $\kappa \in \{0, 0.25, 0.5, 1\}$, 150 simulated studies per cell, each analysed with the planned test: a logistic regression of BET on delay over the EV-matched trials, with standard errors clustered by participant (GEE, exchangeable working correlation), one-sided at $\alpha = 0.05$. The planned regression also included logit(p), but every BET in the final design has $p = 0.5$, so that term is constant and drops out.

Redesign. Each analysis is repeated for a proposed design with delays of 2, 9 and 16 s and 30 EV-matched trials per machine (92 trials in total including the two catch trials), keeping the original stake levels.

5 Results

5.1 Parameter recovery

With 36 trials per participant, recovery is weak for every parameter (Table 1, Figure 1). A quarter of the inverse-temperature estimates run to the upper bound, which happens when a participant's choices are nearly all one option. The redesign improves α, β and b substantially and halves the share of κ fits at a bound, but individual κ estimates remain noisy in both designs. Individual-level κ should not be interpreted on its own.

Table 1: Parameter recovery. Rank correlation between true and recovered values, and the share of fits stuck at a bound.

	Original		Redesign	
Parameter	Rank corr.	At bound	Rank corr.	At bound
α	0.51	10%	0.74	5%
β	0.46	27%	0.60	11%
b	0.44	2%	0.77	0%
κ	0.42	19%	0.48	10%

Table 2: Share of simulated participants for which each criterion selects the true model.

	Original		Redesign	
True model	AIC	BIC	AIC	BIC
EV-RL	80%	93%	83%	96%
AU-RL ($\kappa \in [0.3, 1]$)	42%	23%	66%	50%

5.2 Model recovery

Under the original design, a participant who genuinely savours is classified as AU only 42% of the time by AIC and 23% by BIC (Table 2). The planned per-participant comparison would lean towards "no anticipatory utility" even when it is present. The redesign improves sensitivity to 66% under AIC while keeping specificity above 80%.

5.3 Power

With the planned pilot of 30 participants and $\kappa = 0.5$, the original design detects the effect in 43% of simulated studies (Table 3, Figure 2). Reaching 80% power would take roughly 60 to 120 participants. The redesign reaches 95% with 30 participants and still detects a weaker effect ($\kappa = 0.25$) in 69% of studies.

A single simulated pilot can look far better than the design deserves. One draw of 30 participants at $\kappa = 0.5$ under the original design gave a highly significant slope, while the same design fails in most draws. Power, not one sample, is the right basis for judging a design.

6 Discussion

Why the original design struggles. Two features limit it. First, the delays are close together: at 8, 12 and 16 s the savouring term only doubles from the shortest to the longest machine, so a given κ moves choices little. Spreading the delays to 2, 9 and 16 s gives an eightfold range. Second, 12 choices per machine leave each participant's choice rate very noisy; 30 per machine reduce that noise substantially.

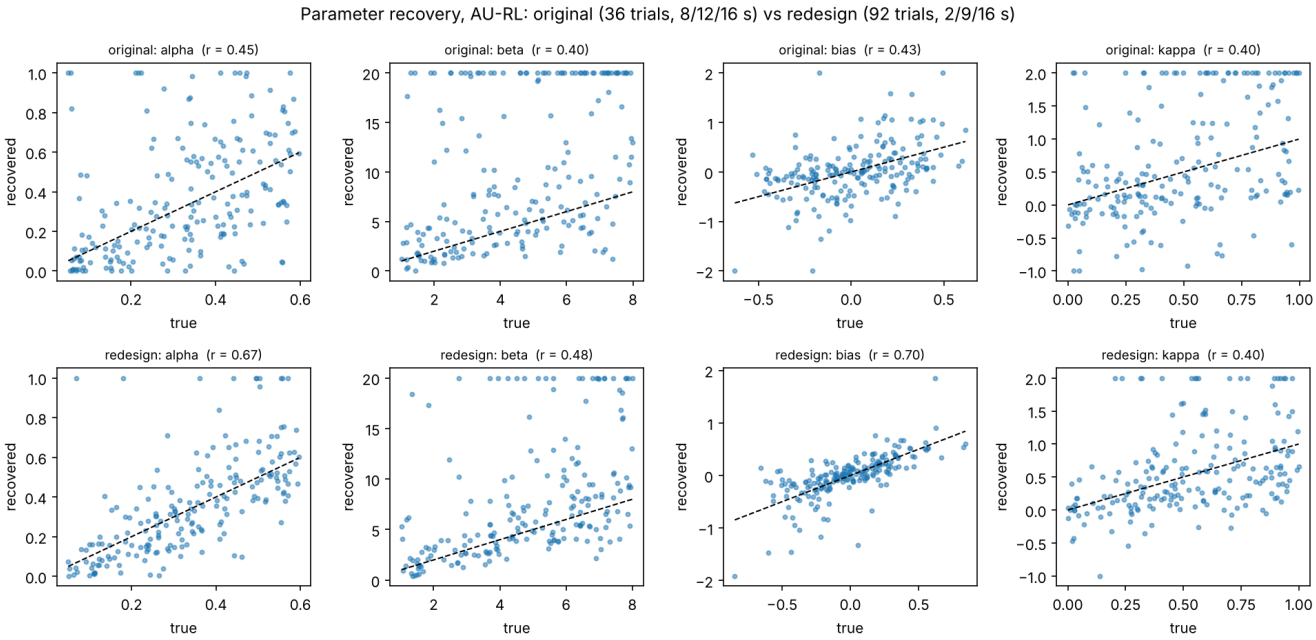


Figure 1: True against recovered parameters for 200 simulated AU participants. Top row: original design. Bottom row: redesign.

Table 3: Share of 150 simulated studies with a significant positive delay effect ($p < 0.05$, one-sided). Rows with $\kappa = 0$ are false-positive rates.

Design	κ	$N = 15$	$N = 30$	$N = 60$	$N = 120$
Original	0	0.10	0.09	0.06	0.04
	0.25	0.25	0.27	0.42	0.65
	0.5	0.31	**0.43**	0.69	0.95
	1.0	0.43	0.61	0.90	0.99
Redesign	0	0.13	0.09	0.07	0.09
	0.25	0.42	0.69	0.89	0.99
	0.5	0.81	**0.95**	1.00	1.00
	1.0	0.95	1.00	1.00	1.00

Recommended analysis. Per-participant model selection under-detects AU at realistic trial counts. The primary test should be group-level: the clustered or mixed-effects regression of choice on delay used here, or a hierarchical fit of κ that pools information across participants.

Limitations.

- The false-positive rate is somewhat above 5% (9–13%) at small N and in the redesign, consistent with the known small-sample bias of GEE sandwich standard errors. A permutation test or bias-corrected sandwich estimator would correct this in a real study.
- Savouring is modelled as linear in delay. If it saturates, as in Iigaya et al. [2016], a wide delay range matters even more.
- True win probabilities equal the prior of 0.5, so the learning rate is only weakly identified by design. This is acceptable for testing AU but limits what the task can say about learning.
- The simulated parameter ranges are assumptions. Pilot data would allow them to be calibrated.
- The prediction-error boost to savouring proposed by Iigaya et al. [2016] is not identifiable here, because every BET has the same win probability. Testing it requires machines that differ in win probability as well as delay, crossed so that the two do not move together.

7 Conclusion

As built, the task is unlikely to detect a moderate anticipatory-utility effect in a 30-person pilot, and per-participant model comparison would mostly miss it. Widening the delay range and increasing trials per machine raises power from 43% to 95% at the same sample size. Simulating the planned analysis before collecting data turns an underpowered pilot into an informative one.

Reproducibility

Code, the task spreadsheet and all results: https://github.com/yamare1/anticipatory-utility-power.

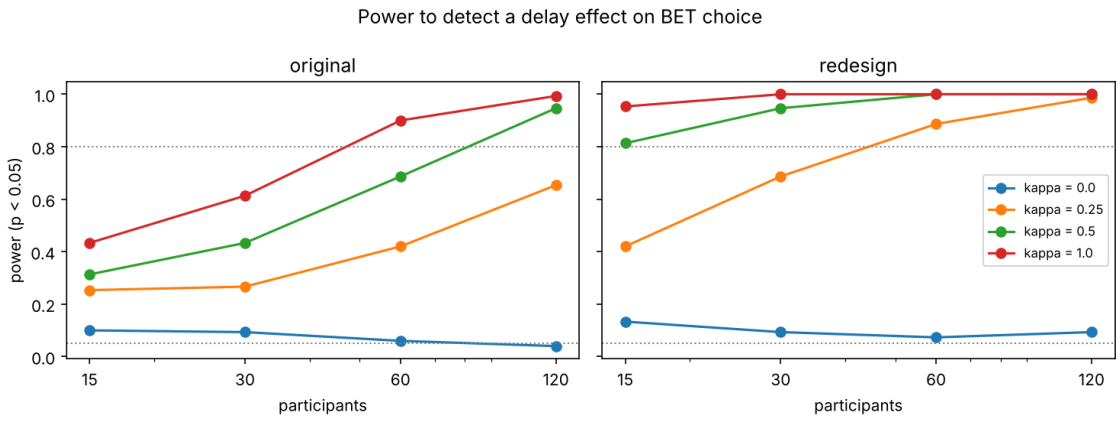


Figure 2: Power of the planned delay test by sample size and true κ. Dotted lines mark 5% and 80%.

All results regenerate from four scripts with fixed random seeds.

References

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